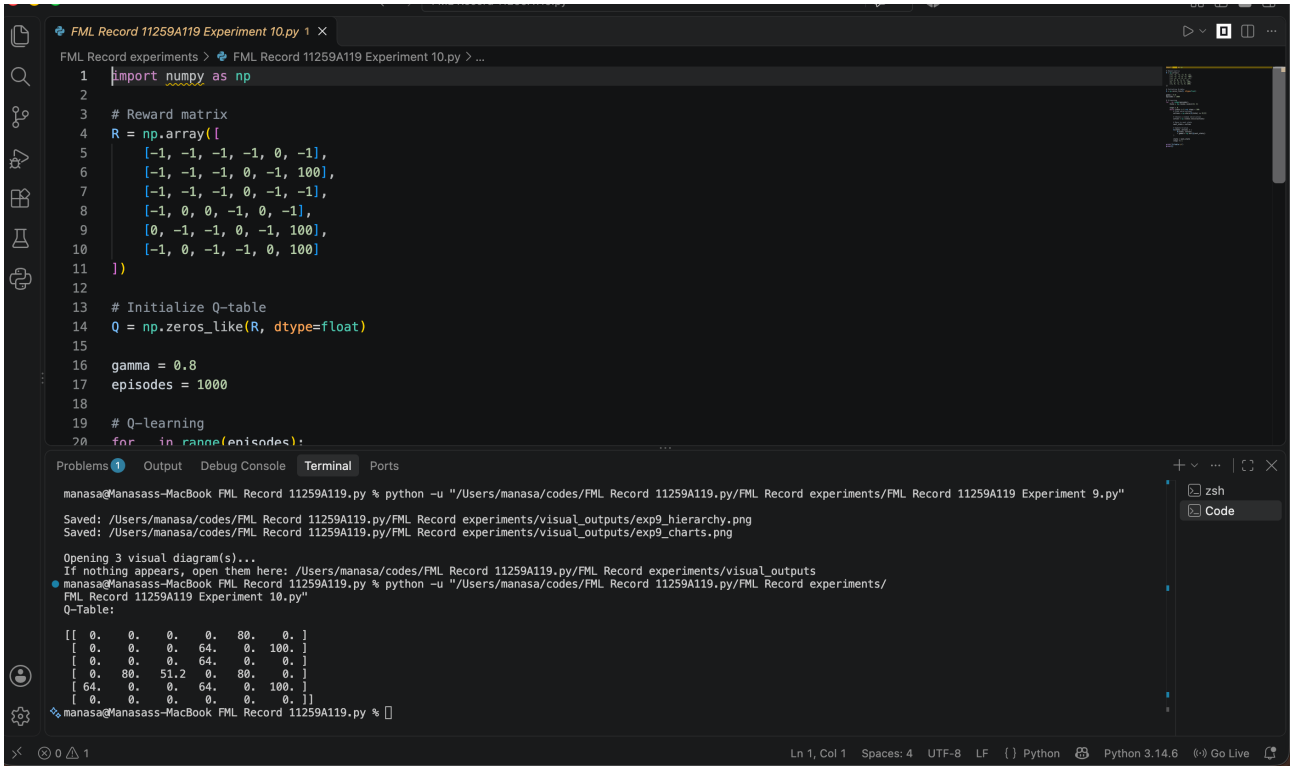
Experiment 10: Basic Reinforcement Learning Using Q-Learning

Aim:

To implement a basic Reinforcement Learning model using the Q-learning algorithm to learn the best actions for different states.

Algorithm:

1. Import the required libraries, including NumPy.
2. Define the states, actions, and reward matrix.
3. Initialize the Q-table with zeros.
4. Set the learning rate, discount factor, and number of episodes.
5. Select actions and obtain rewards from the environment.
6. Update the Q-table using the Q-learning update rule.
7. Repeat the process for the specified number of episodes.
8. Display the final Q-table.



Result:

Thus, the Q-learning algorithm was implemented successfully, and the Q-table was updated through repeated learning episodes to identify rewarding actions.